Decoding "who speaks when" in real-world conversations: Speech detection and speaker identity classification from intracranial EEG

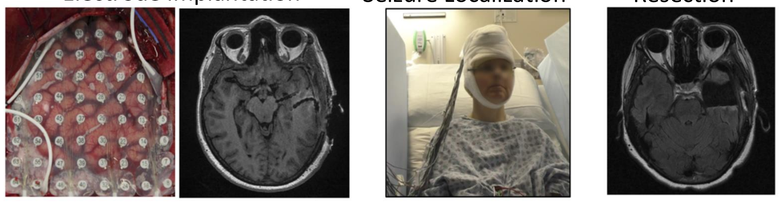
Xinran Yu[1], Arish Alreja[1], Michael J. Ward[2], Taylor Abel[1], Avniel Singh Ghuman[1]
[1]University of Pittsburgh; [2]University of California, Los Angeles

Summary: Using intracranial EEG and eye-tracking glasses in three patients during natural conversation, we decoded when others speak, when the patient speaks, and who is speaking from brain activity. Decoding was above chance in every patient, with strong responses in the superior and transverse temporal cortex. Speech presence and speaker identity can be read out from the human brain in real-world conversation.

Data Acquisition & Analysis

Intracranial EEG

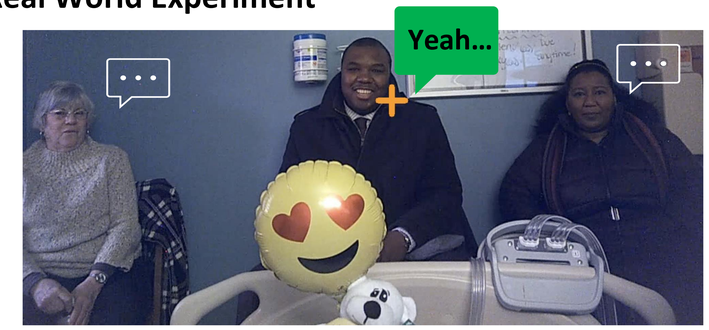
Electrode Implantation | Seizure Localization | Resection



1 – 2 weeks

Participants Consent to Participate in Research Paradigms

Real World Experiment

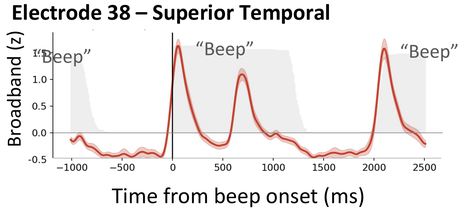


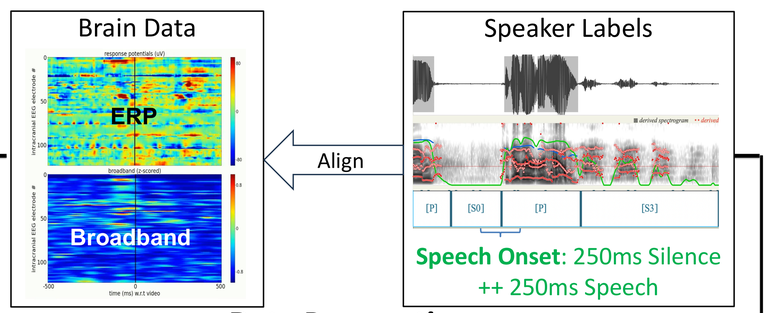
Data

- **3** Patient Subjects
- **~110** intracranial electrodes each
- **~10 Hours** of Real-World Conversation

Sanity Check (Subject A)

Electrode 38 – Superior Temporal





Data Preparation

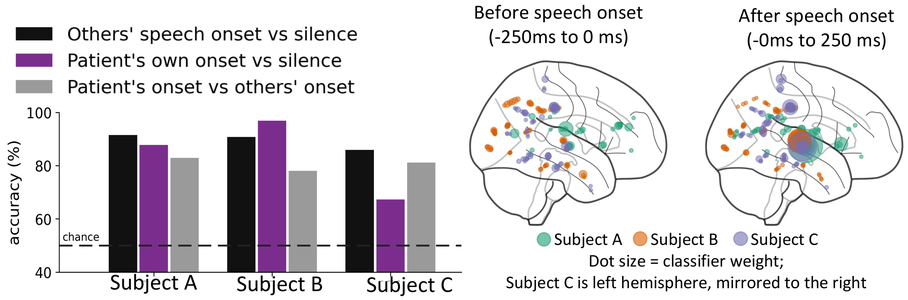
Task

Elastic Net Classifier

$$\min_{\beta_0,\beta}\frac{1}{N}\sum_{i=1}^{N} w_i l(y_i,\beta_0+\beta^T x_i)+\lambda\left[(1-\alpha)\parallel\beta\parallel_2^2/2+\alpha\parallel\beta\parallel_1\right]$$

X → Y

Multi-Subject Analysis



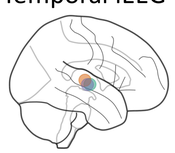
Conclusions

- Others' speech, the patient's own speech, and silence can be distinguished from iEEG in real-world settings in all three patients: 69–81% 3-way accuracy (chance: 33%).
- The classifier uses multiple electrodes, mainly in the superior temporal and parietal cortex.
- A single superior temporal electrode separates two speakers above chance (56–66%): preliminary.
- Next: more speakers and more patients.

Analysis

1. Others, Self, and Silence Classification from iEEG Activity
   - Chance = 33.3%
   - X = All Electrode* [ERP ++ Broadband]
   - Y = {Patient's own speech, All Other Speakers in the room, Silence}

2. Speaker Identity Classification from Superior Temporal iEEG



- Subject A, e38 MNI (46, -14, 2) mm
- Subject B, e48 MNI(60, -20, 7) mm
- Subject C, e64 MNI(57, -17, 2) mm*
  *Left Hemisphere, Mirrored to the right

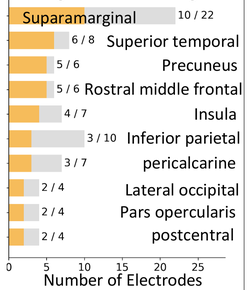
3. Speech-Locked ERP and Broadband Responses (Same Electrode)

Subject A

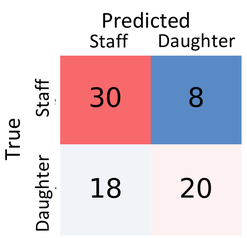


N = 41, F1 = 0.71
Accuracy = 70.7%

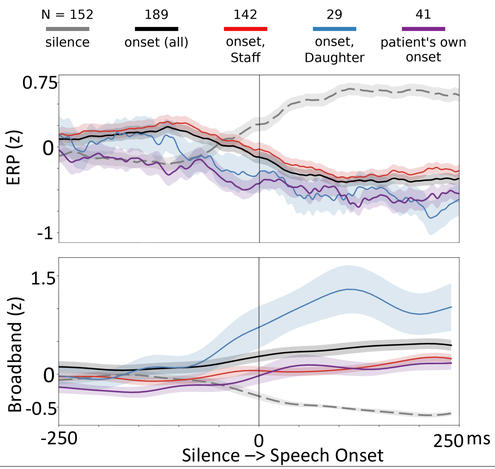
Electrodes Counts by Brain Area
Grey = Implanted
Orange = non-zero weight



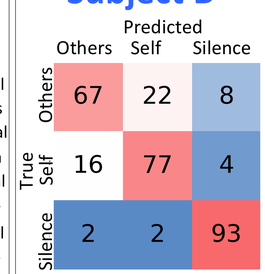
Electrode 38 - **Superior Temporal**



X = [ERP++BB]
Y = Daughter vs Staff
N = 38 per class
Chance = 50%
Accuracy = **65.8%**
F1 = 0.65

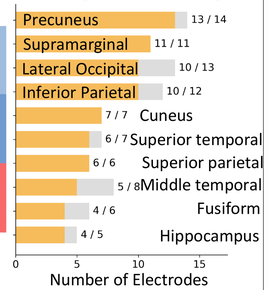


Subject B

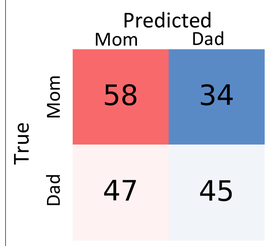


N = 97, F1 = 0.81
Accuracy = 81.4%

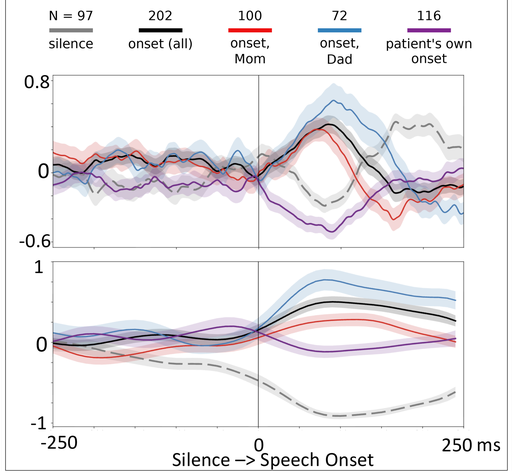
Electrodes Counts by Brain Area
Grey = Implanted
Orange = non-zero weight



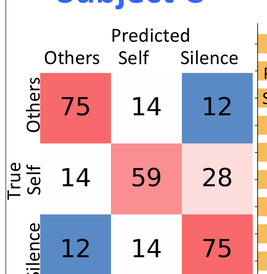
Electrode 48 - **Superior Temporal**



X = [ERP++BB]
Y = Mom vs Dad
N = 92 per class
Chance = 50%
Accuracy = **56.0%**
F1 = 0.56

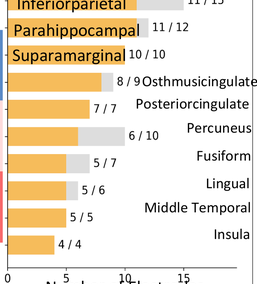


Subject C

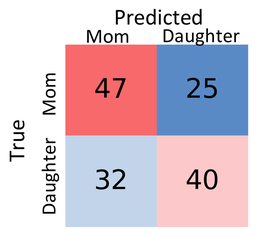


N = 101, F1 = 0.69
Accuracy = 69.0%

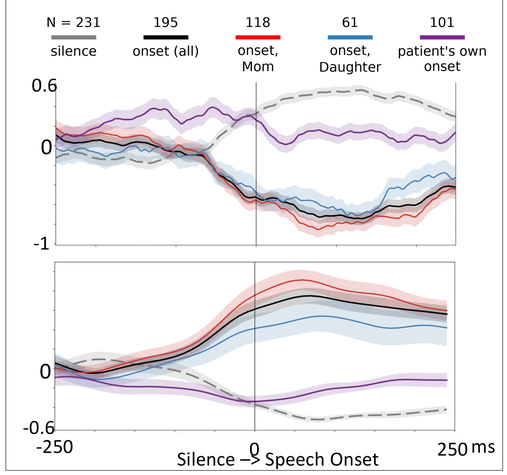
Electrodes Counts by Brain Area
Grey = Implanted
Red = non-zero weight



Electrode 64 - **Superior Temporal**



X = [ERP++BB]
Y = Mom vs Daughter
N = 72 per class
Chance = 50%
Accuracy = **60.4%**
F1 = 0.60



This work was supported by the National Institutes of Health (R01MH132225, R01MH107797, P50MH109429, R90DA060340), the National Science Foundation (1734907), Fredrick Honors College Research Fellowship, Brackenridge Fellowship

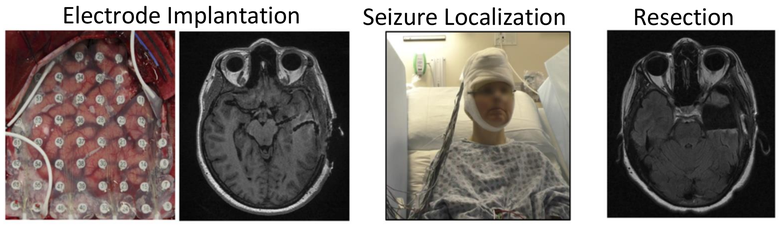
The Neural Correlates of Gaze and Conversation in Real-World Social Interaction Revealed by Intracranial Electroencephalography

Xinran Yu[1], Arish Alreja[1], Michael J. Ward[2], Taylor Abel[1], Avniel Singh Ghuman[1]
[1]University of Pittsburgh; [2]University of California, Los Angeles

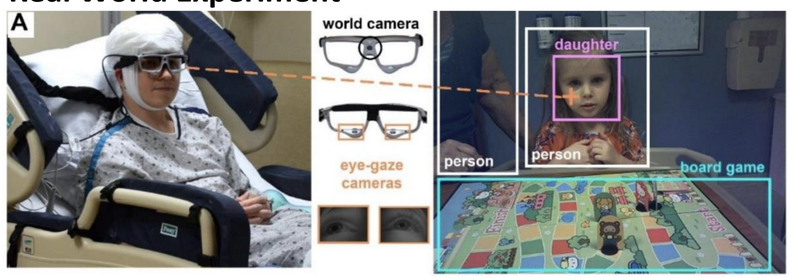
Summary: Using intracranial EEG and eye-tracking glasses in three patients during real-world natural conversation with their visitors, we asked how the brain combines where we look with what we hear. From the 200 ms after each fixation onset, we decoded whether the patient was looking at a face and whether someone was speaking, using broadband activity and ERPs. Neural activity after each fixation carried information about speech in all three patients.

Data Acquisition & Analysis

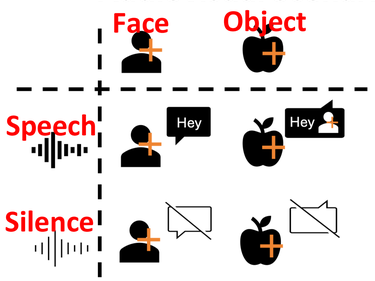
Intracranial EEG



Real World Experiment



Audiovisual Scenario



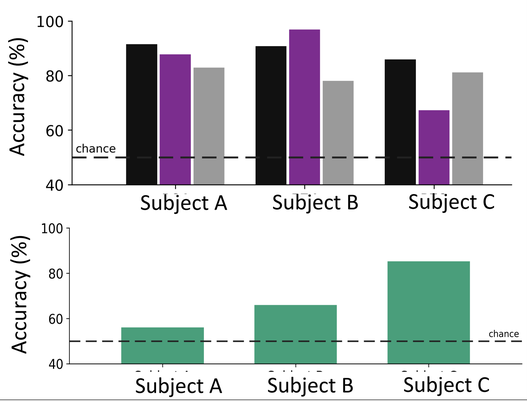
Data

- **3** Patient Subjects
- **~110** intracranial electrodes each
- **~10 Hours** of Real-World Conversation

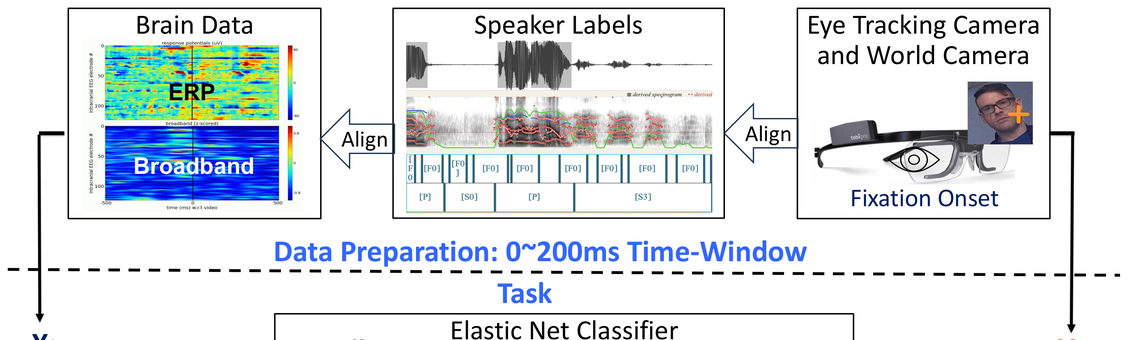
Analysis

1. Face and Speech Classification from iEEG Activity
2. Brain Areas Used by the Classifier
3. Fixation-Locked ERP and Broadband Responses

Feasibility Check: Who Speaks When



Gaze fixation on Face vs Object



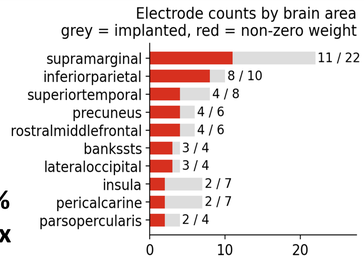
Conclusions

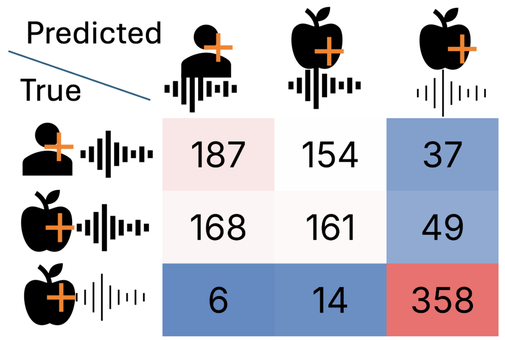
- Gaze target and speech can be read out from the same iEEG signal: 62–67% 3-way accuracy (chance 33%) in all three patients.
- Speech vs. silence is separated best (silence correct in 95%, 89%, 73% of cases); face vs. object during speech is harder.
- The classifier draws on parietal and superior temporal electrodes in every patient.
- Next: control for face–speech overlap, then decode face identity and expression.

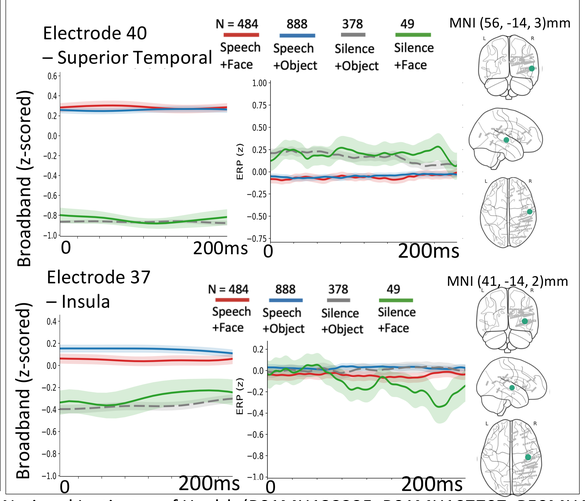
Subject A

110 Electrodes

N = **378 Each**
Chance = **33.3%**
Accuracy = **62.3%**
Confusion Matrix



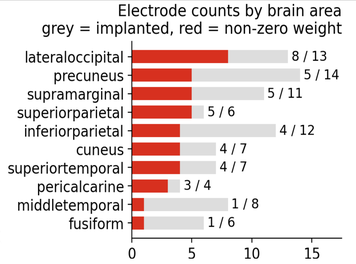


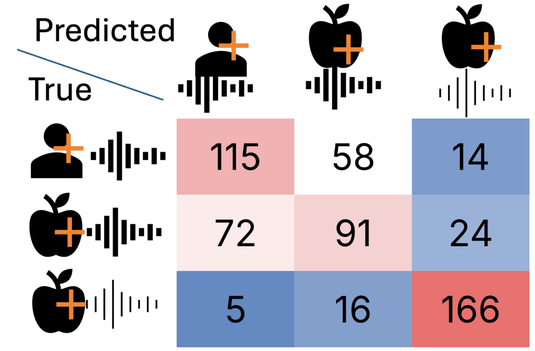


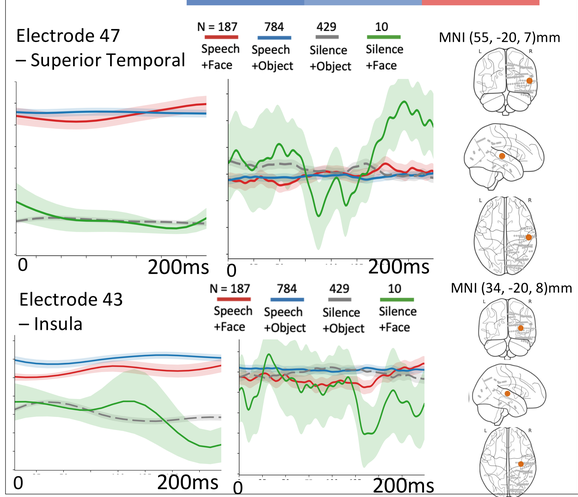
Subject B

109 Electrodes

N = **187 Each**
Chance = **33.3%**
Accuracy = **66.3%**
Confusion Matrix



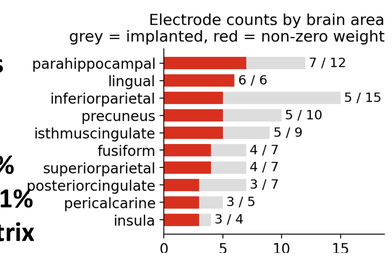


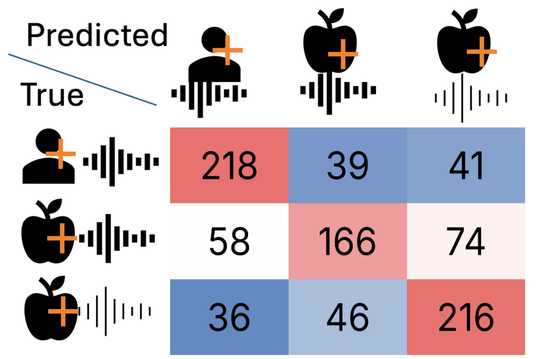


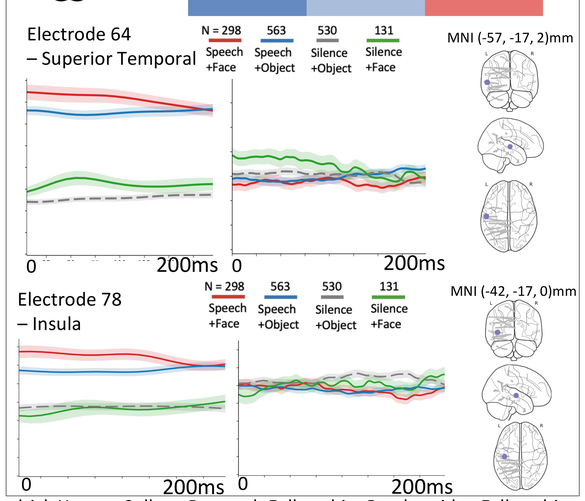
Subject C

124 Electrodes

N = **298 Each**
Chance = **33.3%**
Accuracy = **67.1%**
Confusion Matrix







This work was supported by the National Institutes of Health (R01MH132225, R01MH107797, P50MH109429, R90DA060340), the National Science Foundation (1734907), Fredrick Honors College Research Fellowship, Brackenridge Fellowship